

Imperial College London

BSc/MSci EXAMINATION Summer 2022

*This paper is also taken for the relevant Examination for the Associateship*

## MRES - STATISTICAL METHODS FOR EXPERIMENTAL PHYSICS

**For MLBD MRes Students**

July 6th, 2022: 10:00 to 12:00

*Answer all FOUR questions*

*Marks shown on this paper are indicative of those the Examiners anticipate assigning.*

### **General Instructions**

Complete the front cover of each of the 4 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 4 answer books even if they have not all been used.

**You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.**

1. (i) The binomial distribution describes the probability to observe  $k$  successful outcomes in  $n$  independent trials,

$$P(k; n, p) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

where  $p$  is the probability that a single outcome is successful.

- (a) Show that the expectation value of the binomial distribution is  $np$
- (b) Show that the variance of a binomial distribution is  $np(1-p)$ . You can use the fact that the variance of the sum of independent variables is the same as the sum of the variances of the independent variables.

[10 marks]

- (ii) The Poisson distribution describes the probability to observe  $k$  successful outcomes (events) when those outcomes are rare. Formally, the Poisson distribution arises as the result of a binomial distribution for which  $p \rightarrow 0$  as  $n \rightarrow +\infty$  such that  $\lambda = np$  is constant. The Poisson probability distribution is given by,

$$P(k; \lambda) = \frac{\lambda^k}{k!} e^{-\lambda}$$

- (a) Using the method of moments, find an estimator for the  $\lambda$  parameter  $\hat{\lambda}$  assuming an observed number of events  $k$ .
- (b) Show that the maximum likelihood estimator for  $\lambda$  is the same as the estimator found in the previous question.
- (c) Calculate the third central moment of the Poisson distribution. Remember that the  $n$ -th central moment of the distribution of a discrete random variable  $k$  is defined as,

$$\nu_n = E[(k - E[k])^n],$$

where  $E[\cdot]$  is the expectation value. You may use the following identities;

$$(x^2 + x)e^x = \sum_{k=0}^{+\infty} k^2 \frac{x^k}{k!}$$

$$xe^x = \sum_{k=0}^{+\infty} k \frac{x^k}{k!}$$

$$e^x = \sum_{k=0}^{+\infty} \frac{x^k}{k!}$$

[15 marks]

[Total 25 marks]

2. The Gaussian probability distribution for a continuous random variable  $x$  is given by,

$$f(x; \mu, \sigma) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left( \frac{x - \mu}{\sigma} \right)^2}.$$

- (i) Imagine an experiment that measures observations of  $x$ . Suppose this experiment measures a sequence of  $n$  independent observations  $\{x_1, x_2, \dots, x_n\}$ , assumed to be distributed as Gaussian random variables with the same  $\mu$  and  $\sigma$  parameters.
- (a) Write down the likelihood  $L(\mu, \sigma)$  for the sequence of observations
- (b) Define the quantity  $q = -\ln L(\mu, \sigma)$ . Use this to determine the maximum likelihood estimators  $\hat{\mu}$  and  $\hat{\sigma}$ .

[10 marks]

- (ii) A charged particle tracker is designed to measure the direction of charged particles using 10 layers of silicon detectors. The tracker is designed such that at each layer position ( $x_i$ ) the point at which the particle traverses the silicon along the layer ( $y_i$ ) is measured with some accuracy. Take a look at Figure 1. Each of the data points represents the measured value of the particle position  $y_i$  along the layer  $x_i$ . The measured values  $y_i$  are assumed to be Gaussian random variables with variance  $\sigma^2$ , while the silicon layer positions  $x_i$  are known exactly. The line in the figure represents a hypothesis for the charged particle track, which can be represented by the equation of a straight line,

$$y = mx + c$$

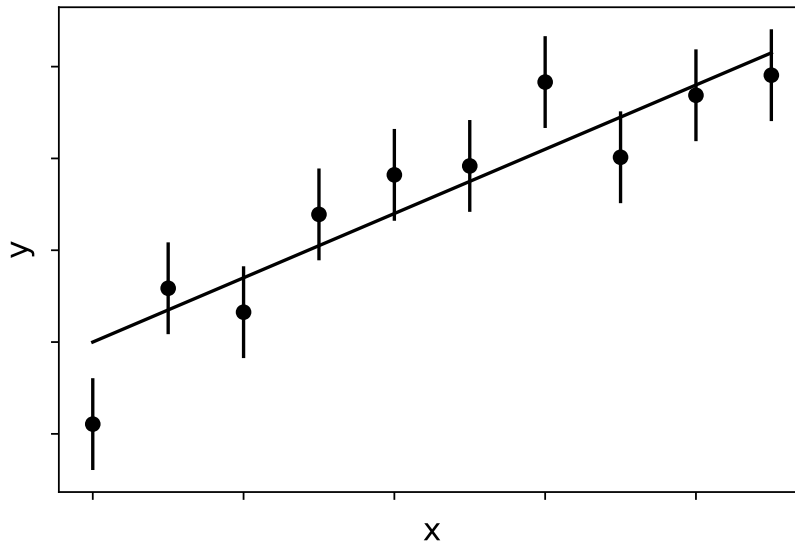


Figure 1

The joint probability density for  $n$  measurements can be expressed as a multivariate Gaussian distribution,

$$f(\underline{\mathbf{y}}; \underline{\boldsymbol{\mu}}, \sigma) = f(\underline{\mathbf{y}}; m\underline{\mathbf{x}} + c, \sigma) = \frac{1}{\sqrt{(2\pi)^n \sigma^{2n}}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (mx_i + c))^2}$$

where in this case  $\underline{\mathbf{y}} = \{y_1, y_2, \dots, y_{10}\}$  and  $n = 10$ .

- Write down the likelihood  $L(m, c)$  and  $q = -\ln L$  for the measured positions as functions of the track parameters  $m$  and  $c$
- Find the maximum likelihood estimators  $\hat{m}$  and  $\hat{c}$
- Describe how you would determine a 68% confidence region for the parameter space  $m$  vs  $c$  from the measurements using the Neyman construction and a profile likelihood ratio test-statistic.
- The quality of the track fit is often thought of in terms of the ‘goodness-of-fit’. The appropriate test statistic is defined as,

$$t = -2 \ln \left( \frac{f(\underline{\mathbf{y}}; \hat{m}, \hat{c}, \sigma)}{f(\underline{\mathbf{y}}; \underline{\mathbf{y}}, \sigma)} \right),$$

where  $f(\underline{\mathbf{y}}; \underline{\mathbf{y}}, \sigma)$  is the ‘saturated model’ for which the mean vector -  $\underline{\boldsymbol{\mu}}$  - of the multivariate Gaussian is equal to the observed values  $\underline{\mathbf{y}}$ . This test statistic is distributed as a chi-square function with 10 degrees of freedom. Write down  $t$  and describe how you would perform a hypothesis test to determine whether the fitted track (the straight line for  $m = \hat{m}$  and  $c = \hat{c}$ ) is a good description of the data.

[15 marks]

[Total 25 marks]

3. (i) (a) In the context of Bayesian parameter inference, and with reference to Bayes' theorem, identify the terms *likelihood*, *prior*, *posterior*, *evidence*.  
 (b) What is the difference between the likelihood and the sampling distribution?

[5 marks]

- (ii) A machine is designed to measure the length  $\mu$  of a crucial component in a medical device. The measurement error of the machine is characterised by a gaussian distribution with zero mean and unknown standard deviation  $\sigma$ . An experiment is performed to infer both  $\mu$  and  $\sigma$ , by making a set  $\{x\}$  of repeated independent measurements of the length:  $x_i; i = 1, \dots, n$ .

- (a) What is the posterior distribution of the model parameters  $\mu$  and  $\sigma$  (up to a normalisation constant which you do not need to calculate)? You should assume that the prior factorises into  $\pi(\mu)\pi(\sigma)$ .  
 (b) Choose a suitable prior  $\pi(\mu)$  and justify your answer.  
 (c) Two possibilities for a prior  $\pi(\sigma)$  suggest themselves:  $\pi(\sigma) \propto \sigma^{-q}$ , with  $q = 0$  and  $q = 1$ .  $q = 0$  is evidently a uniform prior, but provide a justification for the choice  $q = 1$ .

[7 marks]

- (iii) For now, keep  $q$  general.

This experiment is designed to determine the variance of the measurements made by the machine. Show that the marginal posterior distribution of  $\sigma$  is

$$p(\sigma|\{x\}) \propto \frac{1}{\sqrt{n}\sigma^{n+q-1}} \exp\left(-\frac{n\sigma_x^2}{2\sigma^2}\right)$$

where

$$\sigma_x^2 \equiv \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2$$

and  $\bar{x}$  is the average of the  $x_i$ .

[8 marks]

- (iv) On the basis of the marginal distribution of  $\sigma$ , find the most probable value of  $\sigma$ , in terms of  $\sigma_x$ , for the cases  $q = 0$  and  $q = 1$ .

[5 marks]

[Total 25 marks]

4. Two models of the Universe are proposed. In one (Model 0), the Universe is flat, and the curvature of the Universe  $k$  is zero. In the other (Model 1), the curvature  $k$  may be non-zero. A measurement  $d$  of the curvature is made, which is subject to a gaussian error of zero mean and known variance  $\sigma^2$ .

- (i) (a) Write down the likelihood  $p(d|k)$  for Model 0 and for Model 1.  
 (b) In Model 1, assuming a gaussian prior  $\pi(k)$  with zero mean and variance  $\Sigma^2$ , what is the (unnormalised) posterior,  $p(k|d)$ ?

[6 marks]

- (ii) (a) Show that the most probable (maximum *a posteriori*, or MAP) value of  $k$  in Model 1 is

$$k_{\text{MAP}} = \frac{d\Sigma^2}{\sigma^2 + \Sigma^2}.$$

- (b) What is the variance of the posterior in Model 1?

[6 marks]

- (iii) Find an expression for the Bayesian Evidence  $p(d|M_1)$  for Model 1. You may use the result  $\int_{-\infty}^{\infty} \exp(bz - cz^2/2) dz = \sqrt{2\pi/c} \exp[b^2/(2c)]$ . [5 marks]

- (iv) (a) Model 0 has no free parameters. Argue that in this case the Bayesian Evidence  $p(d|M_0)$  is just the likelihood.

- (b) Find an expression for the Bayes factor

$$B_{01} \equiv \frac{p(d|M_0)}{p(d|M_1)}.$$

- (c) How far away does  $d$  have to be from zero for the posterior probability of the two models to be the same? State any further assumptions you make.

[8 marks]

[Total 25 marks]